

Homework 5

Max Points: 25

Directions: Answer each of the questions to the best of your ability. Make sure to show your work in order to receive potential FULL credit. Just stating the answer will result in point deductions (and possibly no credit for the respective problem) and that cannot be disputed through the grade dispute policy. Your answers must typed up and submitted as a PDF. Handwritten solutions will not be accepted. Please note that you can create a separate document when designing your solutions. You do not need to fit your answers in the spaces between each question.

1. (10 points) A ***useless state*** in a pushdown automata is never entered on any input string. Consider the problem of determine whether a pushdown automata has any useless states. Formulate this problem as a language and show that it is **decidable**. Hint: Think about some of properties we learned about context-free grammars.
2. (5 points) In the Silly Post Correspondence Problem, SPCP, the top string in each domino has the same length as the bottom string. Show that SPCP is decidable. Yes decidable, not undecidable.
3. (10 points) Let $T = \{\langle M \rangle \mid M \text{ is a TM that accepts } w^R \text{ whenever it accepts } w\}$. Show that T is **undecidable**.